

HST1 (P53685, *S. cerevisiae*) — Focused Hypothesis Adjudication

Focus: function_assignment · slug `histone-site-specificity-telomeres-and-dna-maintenance`

Source: genes/yeast/HST1/HST1-ai-review.yaml (free-text hypothesis) **Iteration:** 1 of 3

Executive Judgment

Verdict: PARTIALLY SUPPORTED — the compound seed hypothesis is a mix of one well-supported claim and several over-annotated / unresolved claims.

The seed bundles five independent claims. Adjudicated separately:

Sub-claim	Verdict
(1) NAD-dependent deacetylase activity	Supported (direct)
(1b) Specific H3K14 or H3K9 site as the physiological substrate	Weakly supported / unresolved — direct evidence is for <i>general</i> NAD-dependent HDAC activity and for H4K5ac as the in vivo mark; H3K9/H3K14 terms are phylogenetic inference (IBA) only
(2) Participates in native subtelomeric heterochromatin	Refuted for native Hst1 — Hst1 is a <i>gene-specific</i> Sum1–Rfm1 corepressor; regional/telomeric silencing is the Sir2 function
(3) Localizes to telomeric regions	Refuted/unsupported for native Hst1 (annotation is IEA only); telomere/HM localization occurs only in the engineered SUM1-1 gain-of-function context (at HM loci, not telomeres)
(4) Responds to DNA damage	Weakly supported / mostly attributable to other sirtuins — the DNA-damage-response term is IBA only ; the genotoxin-sensitivity phenotype is dominated by Hst3/Hst4 (H3K56ac)
(5) Negatively regulates recombination	Weakly supported / likely refuted for native rDNA recombination — the founding <i>hst1Δ</i> study (P 8810037) reports no rDNA-recombination phenotype ; an IMP annotation (GO:0045950) exists for a specific context, but direct control vs. secondary transcriptional effect is unresolved

Core reasoning. Native Hst1's primary, directly-demonstrated function is **NAD⁺-dependent, Sum1–Rfm1-tethered, promoter-local histone deacetylation that gene-specifically represses** middle-sporulation genes, de novo NAD⁺-biosynthesis (*BNA*) genes, and stress genes. The seed's histone-site, telomere, subtelomeric-heterochromatin, and DNA-damage claims map almost one-to-one onto **IBA/IEA (inferred, non-experimental)** GO annotations that read as paralog carry-over from Sir2. This is the central curation signal.

Most important caveats. (i) "H4K5 preference is not exclusion of other sites" is fair — but the burden of proof for target-specific H3K9/H3K14 native activity is unmet by primary data. (ii)

P 18990212

(H4K16 context effects) and

P 27185881

(conditional HMR effects) were pre-excluded as already-settled and were not re-litigated here.

Evidence Matrix

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Conclusions/ Limitations
12972620 (Bedalov 2003)	Direct assay + expression	Supports (1); qualifies (1b)	NAD-dependent activity; physiological target	Hst1 is an NAD ⁺ sensor; its NAD-dependent deacetylase activity directly represses de novo NAD ⁺ biosynthesis genes	<i>S. cerevisiae</i> , in vivo + transcriptomics	Highly dependent on general repression does not localize activity H3K9me3
17242192 (Mead 2007)	Domain-swap chimeras / mutant	Refutes (2),(3) for native Hst1	Gene-specific vs regional silencing	Hst1→Rfm1/ Sum1 = gene-specific promoter repression; Sir2→Sir3/Sir4 = telomere/HM regional silencing; specificity set by cofactors, not enzymatic difference	<i>S. cerevisiae</i>	Highly clear Hst1 separates
12612074 (McCord 2003)	Interaction + mutant	Qualifies mechanism	Targeting architecture	Rfm1 tethers Hst1 to Sum1 at specific promoters	<i>S. cerevisiae</i>	Highly prominent (not wide)
29066473 (Jaiswal 2017)	Mutant phenotype + ChIP	Qualifies (1b)	In vivo histone substrate	Loss of Sum1–Hst1 → increased H4K5ac at target loci	<i>S. cerevisiae</i> , vegetative	Modulated high H4K5 H3K9 as the documented marker

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11316790 (Rusché & Rine 2001)	Mutant / ChIP	Qualifies (2),(3) — context-gates them	Hst1 in silencing	SUM1-1 relocalizes Sum1 to HM loci via ORC; Hst1 gives Sir-independent silencing w/ nucleosome hypoacetylation	Engineered SUM1-1, HM loci	Highly significant a gap function in natural
11313477 (Sutton 2001)	Mutant / genetic	Qualifies (2),(3)	Hst1 requirement in SUM1-1 silencing	SUM1-1 silencing at HMR requires Hst1 + its NAD ⁺ activity + ORC	Engineered SUM1-1, HMR	Highly consistent dependent
26748095 (Simoneau 2016)	Genome-wide fitness + genetics	Qualifies (4)	Sirtuins & DNA-damage	Genotoxin phenotypes mainly from Hst3/Hst4 (H3K56ac), not Hst1	<i>S. cerevisiae</i>	Highly damaging large Hst1
24710277 (Corbi 2014)	ChIP + genetics	Qualifies (5)	NDT80/meiotic control	Sum1 (not Hst1) determines NDT80 DNA occupancy; Hst1 is the enzymatic partner	<i>S. cerevisiae</i> meiosis	Modeling recombination link trans
26122277 (De Las Peñas 2015, review)	Review/database	Qualifies (2)	Local vs regional silencing	In <i>C. glabrata</i> : Hst1+Sum1– Rfm1 = local silencing; Sir2 = regional/ telomeric	<i>Candida glabrata</i>	Orientation only conserved cross-dividing labo
	Mutant phenotype		rDNA recombination	<i>hst1Δ</i> shows no phenotype in		Highly clear

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8810037 (Derbyshire 1996)	(founding characterization)	Refutes (2),(5) for native Hst1	& regional silencing	rDNA recombination or regional HML α silencing; HST1 is 71% identical to SIR2	<i>S. cerevisiae</i> , native loss-of-function	test against regional
UniProt/GO P53685 (computed)	Database (evidence codes)	Frames all claims	Annotation provenance	H3K14/H3K9/ H4K16 MF, subtelomeric-heterochromatin BP, DNA-damage BP = IBA ; telomeric-region CC = IEA ; only general NAD-HDAC = IDA	Annotation audit	Direct comparison provided (see article)
HST1 vs SIR2 alignment (computed)	Structural/ evolutionary	Qualifies (1b)	Enzymatic-core conservation	Catalytic core 82% identical to Sir2; catalytic His310 conserved — near-identical enzyme, so site "choice" is set by targeting/ cofactors, not intrinsic chemistry	Needleman-Wunsch, UniProt seqs	Direct comparison provided (see article)

GO Curation Implications (leads — require curator verification)

Retain (well-supported): - MF GO:0017136 *histone deacetylase activity, NAD-dependent* — IDA-backed. Keep as the primary MF. - CC GO:0000118 / **Sum1-Rfm1-Hst1 (histone deacetylase complex)**, GO:0005634 **nucleus** — supported by interaction data. - BP: gene-specific negative regulation of transcription; **regulation of NAD⁺ biosynthesis** (well-documented,

);
 (P 12972620
 negative regulation of meiotic/sporulation genes
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Scrutinize / likely too specific (paralog over-annotation): - MF GO:0032041 (**H3K14**) and GO:0046969 (**H3K9**) — **IBA only**. Recommend *do not assert as native experimental MF*; the direct evidence supports only the general NAD-HDAC term, and the one documented in vivo mark is **H4K5ac**. Lead: generalize to GO:0017136, or, if a residue-specific term is desired, the best-evidenced candidate is an H4K5-directed activity (still indirect). GO:0046970 (H4K16) likewise IBA — not native-supported.

Make context-specific / non-core: - BP GO:0031509 **subtelomeric heterochromatin formation** (IBA) and CC GO:0000781 **telomeric region** (IEA) — recommend **remove or down-weight** for native Hst1; these are Sir2 functions. If retained, gate to the SUM1-1 engineered context. - BP GO:0030466 **silent mating-type heterochromatin** (IMP) — retain **only with a SUM1-1 / gain-of-function qualifier**; it is not native targeting.

Treat as unresolved / needs primary check: - BP GO:0006974 **DNA damage response** (IBA) — recommend non-core; phenotype dominated by Hst3/Hst4. - BP GO:0045950 **negative regulation of mitotic recombination** (IMP) — flag for curator: the founding *hst1Δ* study (

(P 8810037
 found **no rDNA-recombination phenotype**, so verify the exact locus/context behind the IMP annotation and whether it reflects direct control vs. secondary transcriptional silencing; recommend treating as **non-core** pending that check.

Avoiding "protein binding": the recommended informative MF is **NAD-dependent histone/protein-lysine deacetylase activity (GO:0017136)**, not a generic binding term.

Mechanistic Scope

Direct gene-product activity: NAD⁺-dependent deacetylation of promoter-proximal nucleosomal histones (in vivo mark: H4K5ac) at loci bound by the DNA-binding protein Sum1, to which Hst1 is tethered by Rfm1. This is a **local, sequence-directed** activity — not domain-wide heterochromatin nucleation/spreading.

Downstream / inferred (not the immediate function): transcriptional repression outputs (middle-sporulation genes; *BNA* NAD-biosynthesis genes; oxidative-stress genes), and any recombination or meiotic-progression phenotypes that follow from that repression. Telomeric/heterochromatic and DNA-damage "functions" are either paralog-inferred (Sir2/Hst3/Hst4) or gain-of-function (SUM1-1) manifestations.

Conflicts and Alternatives

- **Paralog confusion (central issue):** Sir2 (telomere/HM regional silencing, H4K16), Hst3/Hst4 (H3K56ac, genotoxin sensitivity). The seed's telomere/subtelomeric/DNA-damage/site-specificity claims track Sir2/Hst3/Hst4 biology, propagated to Hst1 via IBA.
- **Engineered vs native:** SUM1-1 silencing

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 is at HM loci and is a dominant gain-of-function; it should not be read as native telomere localization.

- **Organism differences:** *C. glabrata* Hst1 does *local* silencing while Sir2 does regional

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P 26122277

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— consistent with the *S. cerevisiae* division of labor, arguing against a native regional/subtelomeric role for Hst1.

- **Enzymatic promiscuity vs physiology:** Hst1's catalytic core is **82% identical to Sir2** (computed; catalytic His310 conserved), so an in-vitro Sir2-family core can deacetylate multiple acetyl-lysines. Per

P 17242192

the divergence is in cofactor/targeting, not enzyme chemistry — hence homology (IBA) transfer of Sir2's H3K9/H3K14/H4K16 activities is plausible in vitro but does **not** establish those as the *major physiological substrate* of native Hst1 (documented in vivo mark: H4K5ac).

Knowledge Gaps

1. **Native histone site-specificity.** Checked: no primary assay pinning native Hst1 to H3K9/H3K14; only general NAD-HDAC (IDA) + in vivo H4K5ac (**P** 29066473). Matters because the seed's site claim is exactly what curators must accept/reject. Resolve with: recombinant Sum1–Rfm1–Hst1 on defined acetyl-histone peptides/nucleosomes, plus locus-specific ChIP of H3K9ac/H3K14ac/H4K5ac in *hst1Δ*.
2. **Direct vs secondary recombination control.** Checked: IMP annotation exists but mechanism unclear. Matters for GO:0045950. Resolve with: recombination assays in *hst1Δ* separating transcriptional-repression loss from a direct chromatin effect (e.g., rDNA/marker recombination with target-gene rescue).
3. **Native telomere occupancy.** Checked: annotation IEA/IBA only; native Sum1 binds specific promoters, not telomeres. Resolve with: genome-wide Hst1 ChIP-seq in wild-type cells to test for any native subtelomeric enrichment.

Discriminating Tests

- **Nucleosome deacetylation assay** of purified Sum1–Rfm1–Hst1 across H3K9ac/H3K14ac/H4K5ac/H4K16ac — distinguishes in-vitro promiscuity from a preferred physiological site.

- **Hst1 ChIP-seq (WT vs SUM1-1)** — directly tests native telomeric/subtelomeric localization vs promoter-specific binding, and isolates the SUM1-1 gain-of-function.
- **Residue-resolved ChIP of acetyl marks in *hst1Δ*** — tests which histone-site acetylation rises when Hst1 is lost (expected: H4K5ac at Sum1 targets).
- **Epistasis for recombination (*hst1Δ* ± restored target-gene expression)** — separates direct recombination control from transcriptional consequence.

Curation Leads (require curator verification)

- **Reference leads (verify snippets):**

- **P 17242192**
— "Hst1 interacts with Rfm1 and Sum1 to repress the transcription of specific middle-sporulation genes. Sir2 interacts with Sir3 and Sir4 to silence genes contained within the silent-mating-type loci and telomere chromosomal regions." → supports gene-specific (not telomeric) native role.

- **P 12972620**
— "The NAD(+)-dependent deacetylase activity of Hst1p represses de novo NAD(+) biosynthesis genes ... suggesting a direct effect." → primary NAD-dependent, gene-specific function.

- **P 29066473**
— "increased H4K5ac at these loci" → documented in vivo histone mark.

- **P 11316790**
11313477 — SUM1-1 gain-of-function context for HM-locus silencing.

- **Candidate GO actions:** keep GO:0017136 (MF, primary) + NAD-biosynthesis/transcription-repression BP + Sum1–Rfm1–Hst1 CC; down-weight/remove IBA site terms (GO:0032041/GO:0046969/GO:0046970), telomere/subtelomeric terms (GO:0000781/GO:0031509), and DNA-damage term (GO:0006974) as non-core; add SUM1-1/engineered qualifier to GO:0030466; flag GO:0045950 for direct-vs-secondary verification.

- **Suggested question to curator:** Is any residue-specific MF (H3K9/H3K14) intended to reflect native physiology or Sir2-family homology? If homology-only, prefer the general NAD-HDAC term.
- **Suggested experiments:** the four Discriminating Tests above.

Artifacts: `HST1_GO_evidence_matrix.csv` (computed GO/evidence-code audit for P53685); `HST1_vs_SIR2_identity.txt` (computed Needleman-Wunsch identity: 82% catalytic-core identity to Sir2, His310 conserved).

Iteration 2 update: Added founding-paper loss-of-function evidence (`P 8810037`):

hst1Δ has no rDNA-recombination or regional HMLα-silencing phenotype), which strengthens the refutation of the native recombination and heterochromatin sub-claims, and added a computed HST1-vs-SIR2 catalytic-core alignment supporting the "targeting-not-enzyme" basis of Hst1/Sir2 divergence.

Iteration 3 update: Targeted search for a primary in-vitro Hst1 histone-site assay returned none, reinforcing that H3K9/H3K14 site-specificity for native Hst1 rests on homology inference rather than direct biochemistry. Consolidated a per-term GO decision table (`HST1_GO_decision_table.csv`) with recommended actions.